

Customer Churn Analysis Dashboard

A Complete Intermediate-Level Project Guide

SQL + Power BI + DAX

Prepared for: Abhishek Rawat | Data Analyst Portfolio Project | 2026

Why This Project?

Before diving in, here is why the Customer Churn Analysis was recommended over the Superstore E-commerce project for your specific profile:

Factor	Superstore (Avoid)	Churn Analysis (Recommended)
Dataset popularity	Extremely overused	Unique, less saturated
Resume overlap	Repeats your retail SQL project	Completely new domain
Interview talking points	Generic sales story	Retention, revenue, segmentation
SQL depth	Moderate	High — window functions, NTILE, CASE
Business story	Moderate	Strong — revenue at risk, root cause
Relevance to your background	Low	Builds on fraud pattern instincts

Dataset

IBM Telco Customer Churn Dataset — Available FREE on Kaggle

Dataset URL:

<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

What the dataset contains:

- 7,043 customer records from a telecom company
- 21 columns covering customer demographics, account info, and service usage
- Key columns: CustomerID, gender, SeniorCitizen, tenure, PhoneService, InternetService, Contract, PaymentMethod, MonthlyCharges, TotalCharges, Churn
- No login required — downloadable directly from Kaggle

Business Problem Statement

"The telecom company is losing customers at an alarming rate. Identify who is churning, why they are leaving, and which customer segments are at highest risk — so the retention team can intervene before revenue is permanently lost."

This framing matters. Write your project title as a question a manager would ask:

"Which customer segments are most at risk of churning, and how much revenue is at stake?"

Tools & Tech Stack

Tool	Purpose	Where Used
MySQL Workbench	Data querying & analysis	Phase 1 — SQL
Power BI Desktop	Dashboard & visualization	Phase 2 — Dashboard
DAX	Calculated measures & KPIs	Phase 2 — Dashboard
GitHub	Portfolio hosting	Final delivery
Power BI Service	Publishing dashboard	Final delivery

Phase 1 — SQL Analysis (MySQL Workbench)

Duration: 3-4 days | Skill level demonstrated: Intermediate

Step 1: Database Setup

Create the schema and import the CSV dataset into MySQL Workbench:

```
-- Create database
CREATE DATABASE churn_analysis;
USE churn_analysis;

-- After importing CSV via Import Wizard, verify:
SELECT COUNT(*) FROM customers;    -- Should return 7043
SELECT * FROM customers LIMIT 5;
```

Query 1 — Overall Churn Rate

The first and most important KPI — what percentage of customers are leaving?

```
SELECT
  COUNT(*) AS Total_Customers,
  SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS Churned_Customers,
  SUM(CASE WHEN Churn = 'No' THEN 1 ELSE 0 END) AS Retained_Customers,
  ROUND(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2)
  AS Churn_Rate_Pct
FROM customers;
```

```
-- Expected output: ~26.5% churn rate
```

Query 2 — Churn by Contract Type

Month-to-month contracts are the biggest risk factor. This query proves it:

```
SELECT
    Contract,
    COUNT(*) AS Total_Customers,
    SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS Churned,
    ROUND(AVG(CASE WHEN Churn = 'Yes' THEN 1.0 ELSE 0 END) * 100, 2) AS Churn_Pct
FROM customers
GROUP BY Contract
ORDER BY Churn_Pct DESC;

-- Expected: Month-to-month ~42%, One year ~11%, Two year ~3%
```

Query 3 — Average Charges: Churned vs Retained

Do churned customers pay more? This reveals pricing pressure as a driver:

```
SELECT
    Churn,
    ROUND(AVG(MonthlyCharges), 2) AS Avg_Monthly_Charge,
    ROUND(AVG(tenure), 1) AS Avg_Tenure_Months,
    ROUND(AVG(TotalCharges), 2) AS Avg_Total_Revenue
FROM customers
GROUP BY Churn;
```

Query 4 — Revenue Lost Due to Churn

Quantify the financial impact — this is the number that goes in your resume bullet:

```
SELECT
    ROUND(SUM(CASE WHEN Churn='Yes' THEN MonthlyCharges ELSE 0 END), 2)
    AS Monthly_Revenue_Lost,
    ROUND(SUM(CASE WHEN Churn='Yes' THEN TotalCharges ELSE 0 END), 2)
    AS Total_Revenue_Lost,
    ROUND(SUM(CASE WHEN Churn='No' AND Contract='Month-to-month'
    THEN MonthlyCharges ELSE 0 END), 2)
    AS Revenue_At_Risk
FROM customers;
```

Query 5 — Churn by Payment Method

Electronic check users churn the most. This drives a payment nudge recommendation:

```

SELECT
    PaymentMethod,
    COUNT(*) AS Total,
    ROUND(AVG(CASE WHEN Churn='Yes' THEN 1.0 ELSE 0 END) * 100, 2) AS Churn_Pct
FROM customers
GROUP BY PaymentMethod
ORDER BY Churn_Pct DESC;

```

Query 6 — High-Risk Customer Segmentation (Window Function)

Use NTILE to bucket retained customers by tenure and charges — find who is most at risk of churning next:

```

SELECT
    customerID,
    tenure,
    MonthlyCharges,
    Contract,
    PaymentMethod,
    NTILE(4) OVER (ORDER BY MonthlyCharges DESC) AS Charge_Quartile,
    NTILE(4) OVER (ORDER BY tenure ASC) AS Tenure_Quartile,
    CASE
        WHEN NTILE(4) OVER (ORDER BY tenure ASC) = 1
            AND NTILE(4) OVER (ORDER BY MonthlyCharges DESC) = 1
        THEN 'HIGH RISK'
        ELSE 'MONITOR'
    END AS Risk_Flag
FROM customers
WHERE Churn = 'No' AND Contract = 'Month-to-month';

```

Query 7 — Senior Citizen Churn Analysis

An insight specific to this dataset — senior citizens churn at nearly double the rate:

```

SELECT
    CASE WHEN SeniorCitizen = 1 THEN 'Senior' ELSE 'Non-Senior' END AS Segment,
    COUNT(*) AS Total,
    ROUND(AVG(CASE WHEN Churn='Yes' THEN 1.0 ELSE 0 END)*100, 2) AS Churn_Pct,
    ROUND(AVG(MonthlyCharges), 2) AS Avg_Monthly_Charge
FROM customers
GROUP BY SeniorCitizen;

```

Phase 2 — Power BI Dashboard

Duration: 3-4 days | Output: 3-page interactive dashboard published to Power BI Service

Data Import Steps

1. Open Power BI Desktop
2. Home > Get Data > Text/CSV > select the telco CSV file
3. In Power Query Editor: set correct data types (TotalCharges column has blanks — replace with 0, then set to Decimal)
4. Create a Date table using DAX for time intelligence (optional but impressive)
5. Click Close & Apply

Page 1 — Executive Churn Overview

The landing page. A manager should understand the situation in under 30 seconds.

Visual	What It Shows	Visual Type
Churn Rate %	Overall headline KPI	KPI Card
Total Churned Customers	Raw count	KPI Card
Monthly Revenue Lost	Financial impact	KPI Card
Avg Tenure of Churned	How long before they leave	KPI Card
Churned vs Retained	Proportion overview	Donut Chart
Churn by Contract Type	Biggest driver	Clustered Bar Chart
Churn by Internet Service	Service-level breakdown	Stacked Bar Chart

Page 2 — Customer Risk Segmentation

The analytical deep-dive page. Shows WHO is at risk and what they look like.

Visual	What It Shows	Visual Type
Tenure vs Monthly Charges	Churn pattern by value & loyalty	Scatter Plot (colour by Churn)
Churn by Payment Method	Electronic check risk signal	Horizontal Bar Chart
Churn by Senior Citizen	Demographic segment risk	Clustered Column Chart
High-Risk Customer Table	Top 20 retained at-risk customers	Table with conditional formatting
Slicers	Filter by Contract, Gender, Service	Slicer panel

Page 3 — Revenue Impact & Recommendations

The business action page. Converts analysis into actionable recommendations.

Visual	What It Shows	Visual Type
Revenue Lost vs Retained	Financial storytelling	Waterfall Chart
Revenue at Risk (Month-to-month)	What could still be saved	Gauge or KPI Card
Churn by Tenure Bucket	When customers leave (0-12, 13-24, 25+)	Line/Area Chart

Top Recommendations	Action items for retention team	Text Card / Shape
---------------------	---------------------------------	-------------------

DAX Measures to Write

Write these in the Modeling tab > New Measure:

Measure 1 — Churn Rate %

```
Churn Rate % =
DIVIDE(
    CALCULATE(COUNTROWS(customers), customers[Churn] = "Yes"),
    COUNTROWS(customers)
) * 100
```

Measure 2 — Monthly Revenue Lost

```
Monthly Revenue Lost =
CALCULATE(
    SUM(customers[MonthlyCharges]),
    customers[Churn] = "Yes"
)
```

Measure 3 — Revenue at Risk

```
Revenue at Risk =
CALCULATE(
    SUM(customers[MonthlyCharges]),
    customers[Churn] = "No",
    customers[Contract] = "Month-to-month"
)
```

Measure 4 — Average Tenure (Churned Only)

```
Avg Tenure Churned =
CALCULATE(
    AVERAGE(customers[tenure]),
    customers[Churn] = "Yes"
)
```

Key Insights to Highlight (Pre-Answered for You)

These are the findings you should surface in your dashboard and GitHub README:

- Overall churn rate is approximately 26.5% — 1 in 4 customers is leaving
- Month-to-month contract holders churn at ~42% vs only 3% for two-year contracts
- Electronic check users have the highest churn rate among payment methods
- Senior citizens churn at nearly double the rate of non-senior customers

- Churned customers have significantly shorter tenure (avg ~18 months vs ~37 months for retained)
- Customers with higher monthly charges and shorter tenure are the highest-risk retained segment

Ready-to-Use Resume Bullet

Customer Churn Analysis Dashboard | SQL · Power BI · DAX | GitHub / Live Demo

- Analysed 7,000+ telecom customer records using MySQL — identifying contract type and payment method as primary churn drivers, with month-to-month customers churning at 42% vs 3% for two-year contracts.
- Built a 3-page interactive Power BI dashboard with DAX measures quantifying revenue lost and revenue at risk, enabling targeted retention strategies for high-risk customer segments.
- Applied window functions (NTILE) to segment retained customers by churn risk, surfacing a prioritised list of 20+ high-risk accounts for immediate retention outreach.

GitHub Repository Structure

A clean GitHub repo is part of the project. Structure it like this:

```
customer-churn-analysis/  
  README.md           <- Business problem, key findings, dashboard link  
  /data  
    WA_Fn-UseC_-Telco-Customer-Churn.csv  
  /sql  
    01_churn_rate.sql  
    02_churn_by_contract.sql  
    03_avg_charges.sql  
    04_revenue_lost.sql  
    05_payment_method.sql  
    06_risk_segmentation.sql  
    07_senior_citizen.sql  
  /powerbi  
    churn_dashboard.pbix  
  /screenshots  
    page1_overview.png  
    page2_risk.png  
    page3_revenue.png
```

Suggested 10-Day Timeline

Days	Task	Output
Day 1	Download dataset, set up MySQL, import data	Schema ready
Day 2	Write SQL Queries 1-3 (Churn rate, Contract, Charges)	3 queries done

Day 3	Write SQL Queries 4-7 (Revenue, Payment, NTILE, Senior)	All 7 queries done
Day 4	Import into Power BI, clean in Power Query	Clean data model
Day 5-6	Build Page 1 — Executive Overview	Page 1 complete
Day 7-8	Build Pages 2 & 3 — Risk & Revenue	All 3 pages done
Day 9	Write DAX measures, add slicers, format dashboard	Polished dashboard
Day 10	Push to GitHub, publish to Power BI Service, update resume	Portfolio ready

Final Note

This project demonstrates: SQL depth (window functions, CASE, NTILE), Power BI storytelling (3-page dashboard), DAX measures, business thinking (revenue quantification), and GitHub documentation. That is the complete package hiring managers look for in an entry-level Data Analyst.

Good luck, Abhishek. Build this, push it to GitHub, publish the dashboard, and add the resume bullet. That is one strong portfolio piece done.